

Express.js Theory Guide (Quick Revision)

1. Routing

Routing defines how your server responds to different URLs and HTTP methods (GET, POST, PUT, DELETE). In Express, routes are created using `app.get()`, `app.post()`, etc. Each route handles a specific endpoint.

2. Request & Response

`req` (request) contains data sent by the client (body, params, query). `res` (response) is used to send data back (`res.send()`, `res.json()`, `res.status()`).

3. Middleware

Middleware are functions that run before reaching the route. They can modify `req/res` or stop the request. Examples: logging, authentication, parsing JSON.

4. CRUD Operations

CRUD = Create, Read, Update, Delete. POST → Create, GET → Read, PUT/PATCH → Update, DELETE → Delete. Used to manage database data.

5. Template Engines

Used to render dynamic HTML from the backend (e.g., EJS, Pug). They combine backend data with frontend templates.

6. Error Handling

Handle errors using middleware with (`err`, `req`, `res`, `next`). Always send proper status codes like 404 or 500.

7. 3rd Party NPM Packages

Express apps often use packages like `dotenv`, `mongoose`, `cors`, `bcrypt`, `jsonwebtoken`. Install using `npm install`.

8. Controllers

Controllers separate logic from routes. Routes handle URL, controllers handle business logic.

9. Fetching From Frontend

Frontend uses fetch or axios to send HTTP requests to Express backend. Example:
`fetch('/api/users').then(res => res.json())`.

10. Environment Variables

Used to store sensitive data (PORT, DB_URL). Managed using dotenv and .env file.